

Introduction to Neural Networks

Transfer learning

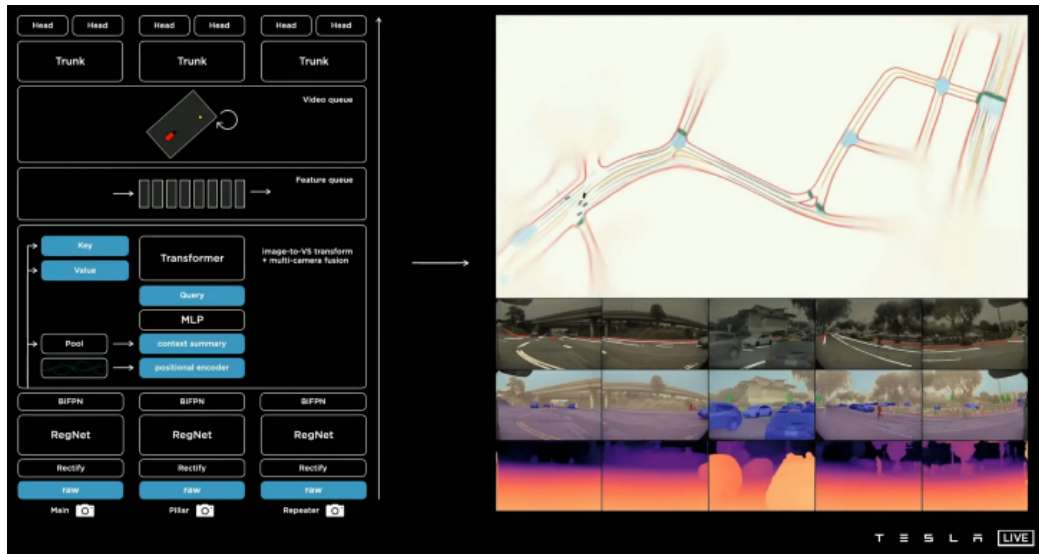
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Neural networks advances



The reasons of success

Dataset sizes

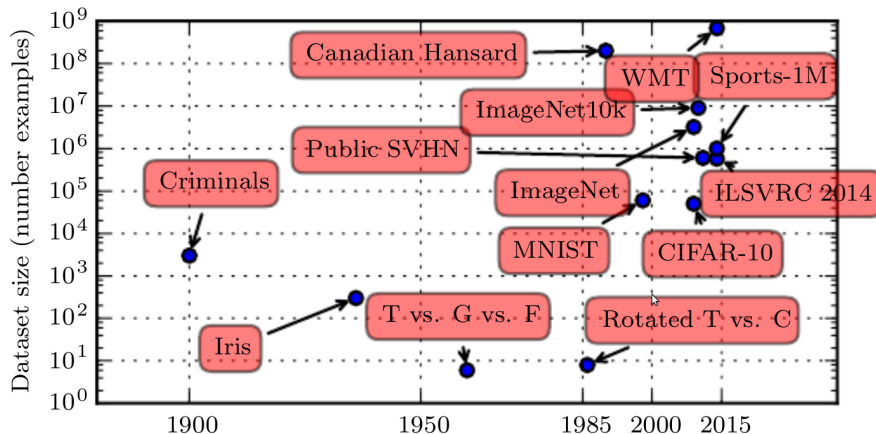


Figure: Dataset sizes over time

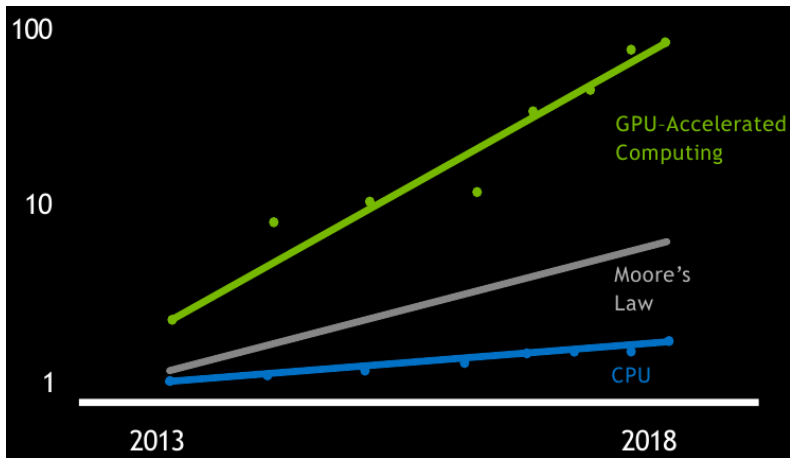
The reasons of success

Dataset sizes

- <https://image-net.org>
- <https://neurohive.io/en/datasets/tencent-dataset/>
- <http://xviewdataset.org/>

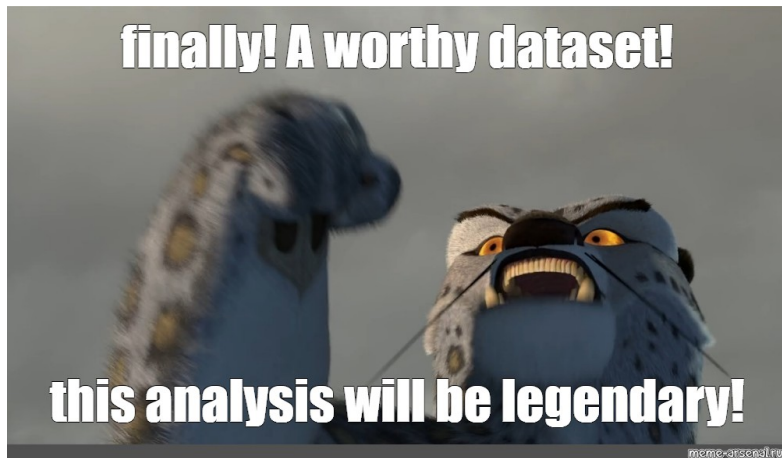
The reasons of success

GPU processing times - Is it enough?



Practical problems

- It is difficult to get data and computing power.



Transfer Learning

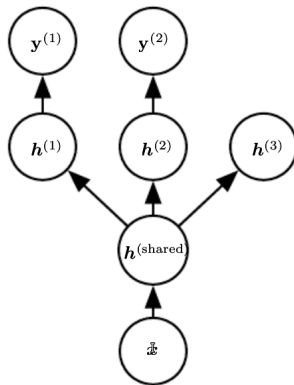
"What has been learned in one sitting is exploited to improve generalization in another sitting"



I know kung fu.

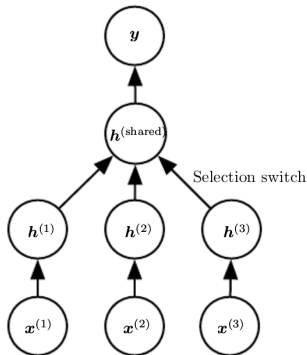
Transfer Learning

- Many visual categories share low level notions.

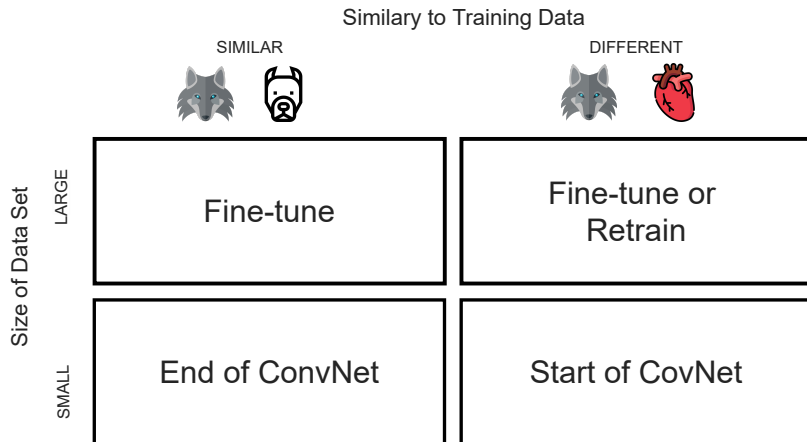


Transfer Learning

- In some cases, the knowledge shared in on the top layers. e.g. speech recognition.

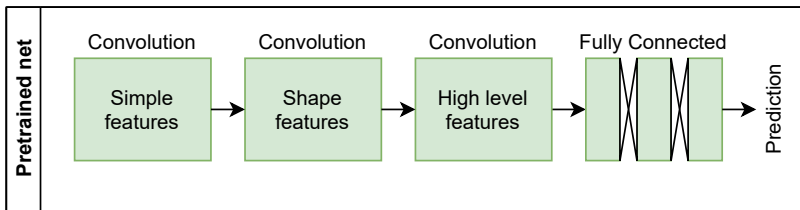


The Four Main Cases When Using Transfer Learning



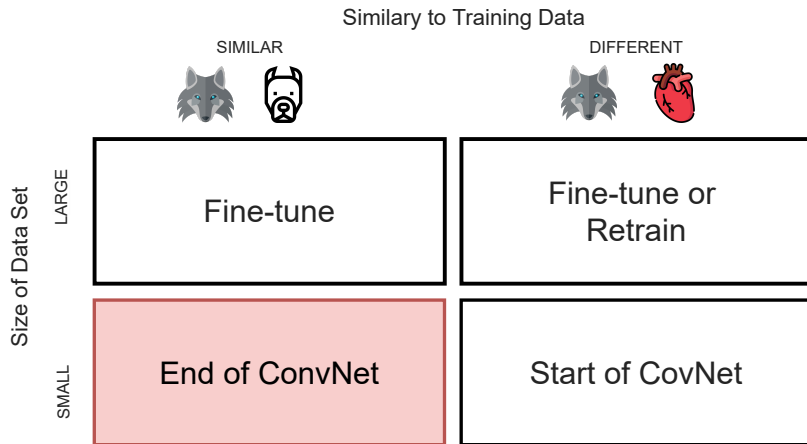
Demonstration network

Pre-trained convolutional neural network



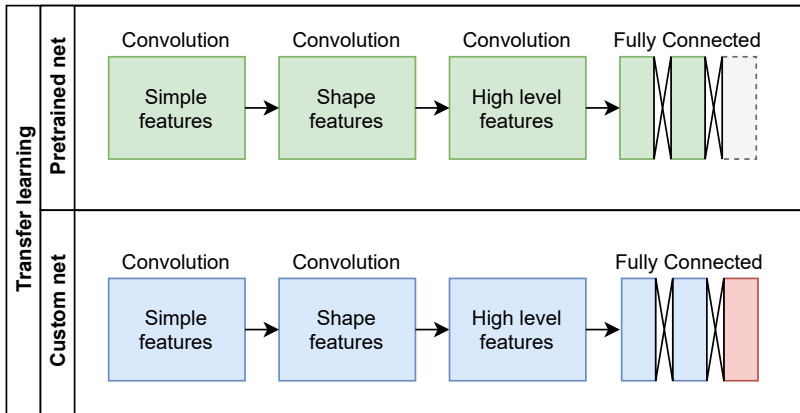
Case 1: Small Data Set, Similar Data

Guide for How to Use Transfer Learning



Case 1: Small Data Set, Similar Data

Guide for How to Use Transfer Learning



Case 1: Small Data Set, Similar Data

Guide for How to Use Transfer Learning

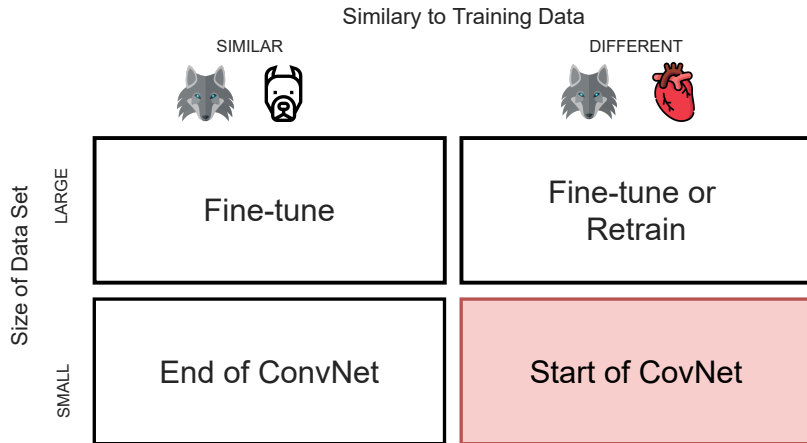
If the new data set is small and similar to the original training data:

- slice off the end of the neural network
- add a new fully connected layer that matches the number of classes in the new data set
- randomize the weights of the new fully connected layer; freeze all the weights from the pre-trained network
- train the network to update the weights of the new fully connected layer

To avoid overfitting on the small data set, the weights of the original network will be held constant rather than re-training the weights.

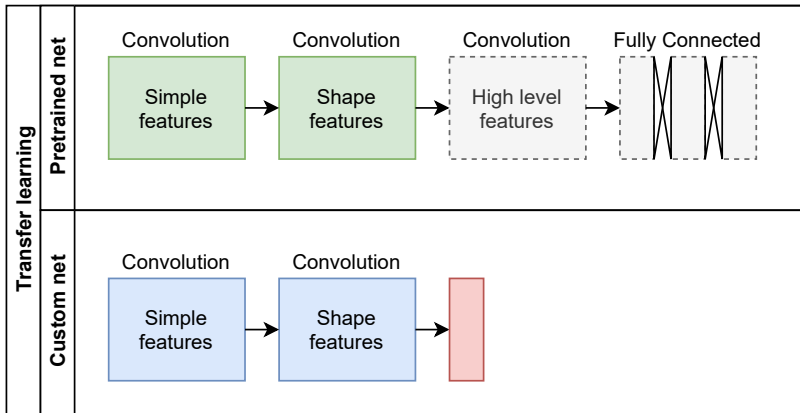
Case 2: Small Data Set, Different Data

Guide for How to Use Transfer Learning



Case 2: Small Data Set, Different Data

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Case 2: Small Data Set, Different Data

Guide for How to Use Transfer Learning

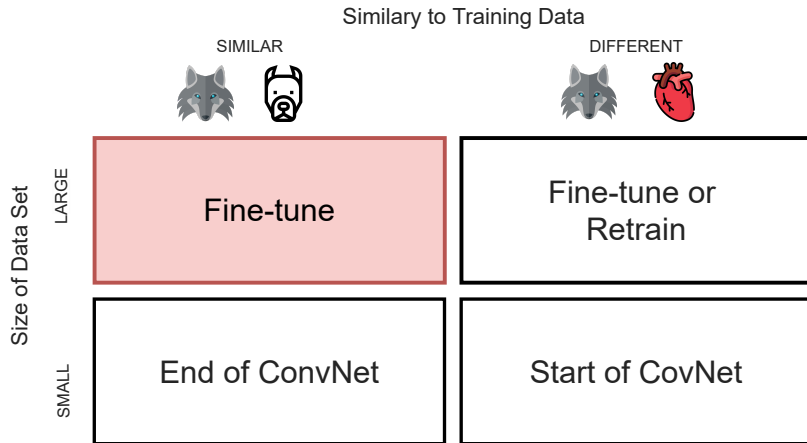
If the new data set is small and different from the original training data:

- slice off most of the pre-trained layers near the beginning of the network
- add to the remaining pre-trained layers a new fully connected layer that matches the number of classes in the new data set
- randomize the weights of the new fully connected layer; freeze all the weights from the pre-trained network
- train the network to update the weights of the new fully connected layer

Because the data set is small, overfitting is still a concern. To combat overfitting, the weights of the original neural network will be held constant, like in the first case.

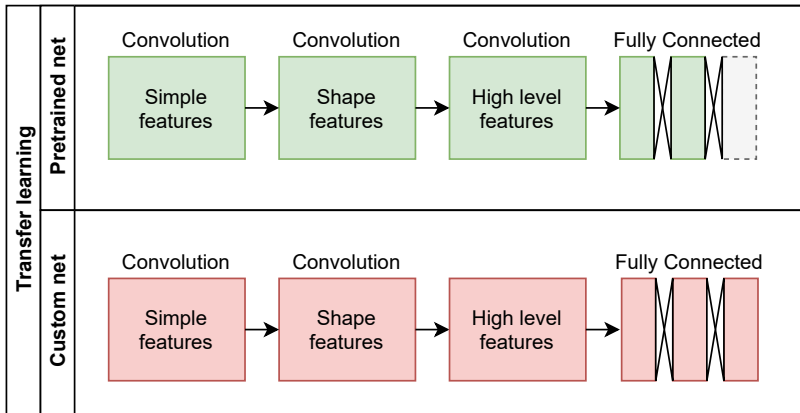
Case 3: Large Data Set, Similar Data

Guide for How to Use Transfer Learning



Case 3: Large Data Set, Similar Data

Guide for How to Use Transfer Learning



Case 3: Large Data Set, Similar Data

Guide for How to Use Transfer Learning

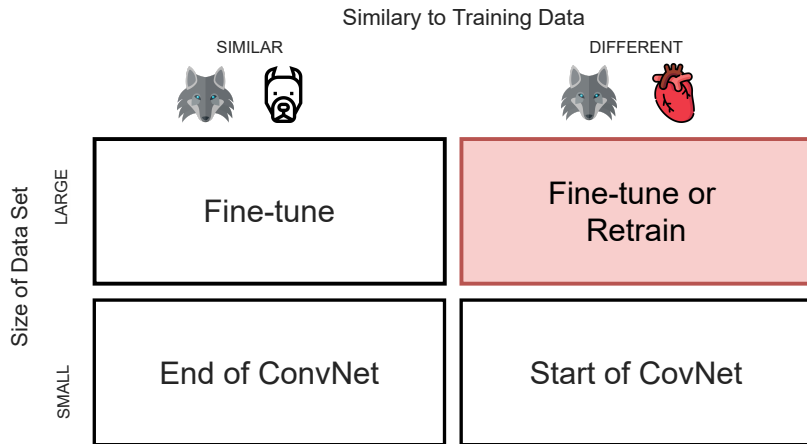
If the new data set is large and similar to the original training data:

- remove the last fully connected layer and replace with a layer matching the number of classes in the new data set
- randomly initialize the weights in the new fully connected layer
- initialize the rest of the weights using the pre-trained weights
- re-train the entire neural network

Overfitting is not as much of a concern when training on a large data set; therefore, you can re-train all of the weights.

Case 4: Large Data Set, Different Data

Guide for How to Use Transfer Learning



Case 4: Large Data Set, Different Data

Guide for How to Use Transfer Learning

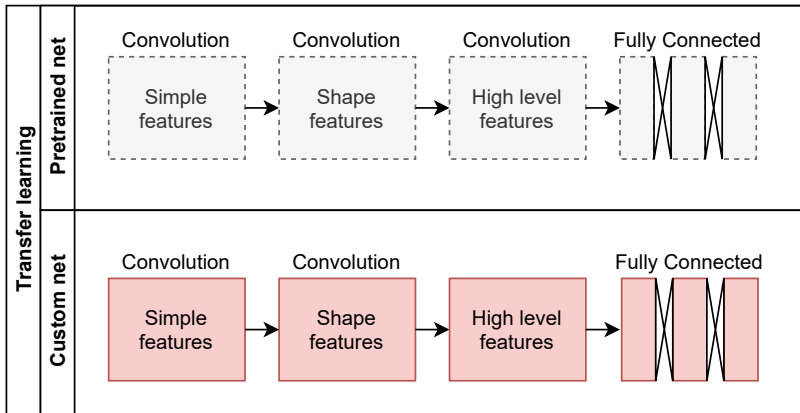
If the new data set is large and different from the original training data:

- remove the last fully connected layer and replace with a layer matching the number of classes in the new data set
- retrain the network from scratch with randomly initialized weights
- alternatively, you could just use the same strategy as the "large and similar" data case

Even though the data set is different from the training data, initializing the weights from the pre-trained network might make training faster. So this case is exactly the same as the case with a large, similar data set.

Case 4: Large Data Set, Different Data

Guide for How to Use Transfer Learning



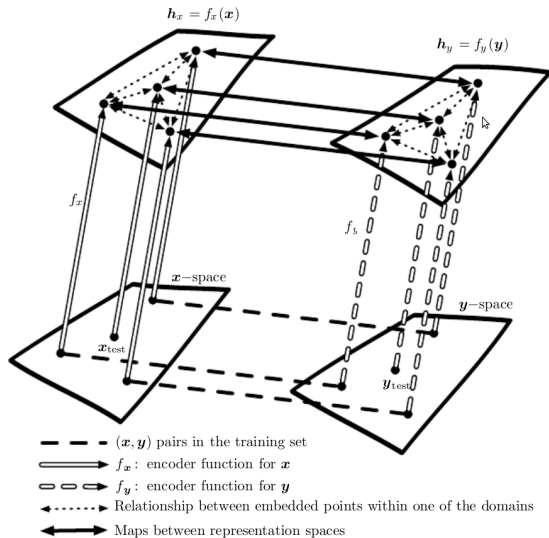
Transfer learning be like



One-shot learning

- Only one labeled example of the transfer task is given for one-shot learning.
- E.g. consider the problem of having a learner read a large collection of text and then solve object recognition problems.
- The model is trained to estimate the conditional distribution $p(y|x, T)$. where T is a random variable describing the task.

One-shot learning



- [Goodfellow](#) Goodfellow, I., Bengio, Y., Courville, A., & Bengio, Y. (2016). Deep learning (Vol. 1). Cambridge: MIT press.
- [Sugriva](#) 15 Sucar, Luis Enrique. Probabilistic Graphical Models: Principles and Applications. Springer, 2015.
- [Smola](#) Alex Smola and S.V.N. Vishwanathan, Introduction to Machine Learning, Cambridge University Press.
- [University of California](#) Self Driving Car Nanodegree